

Certifying Regime Detectors Before Use

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Abstract

Value-at-risk forecasts are backtested before they are trusted; regime detectors, whose labels and triggers enter inference and allocation just as directly, face no analogous pre-use validation. We propose one. A certification protocol treats a frozen detector as a measurement instrument and, for a single pre-declared claim, measures five operating characteristics: power against a declared signal family (by controlled signal injection), false-alarm behavior under dependence-preserving resampled nulls, equivalence of outputs across calendar, volume, and event-time bars within a preregistered margin, detector-output information about the declared target, and the net value of that information under a declared cost convention. The protocol returns a certificate with a disposition and a gate-level failure profile, and refuses comparisons between detectors whose targets differ.

We first certify a Holm-corrected variance-ratio cascade on BTCUSDT one-minute perpetual futures (2021–2025; $n = 2,415,148$ bars) and withhold the verdict, for measured reasons: the cascade cannot detect variance-ratio departures ≤ 0.10 anywhere on the preregistered 96-cell grid; its nominal 5% rule produced zero false alarms in 100 null replicates, a size distortion traceable to a mis-centered asymptotic reference; cross-clock equivalence certifies at $q=2$ but fails at the primary horizon $q=5$; and trigger information has a net value whose sign depends on the cost-attribution convention. We then show the diagnosis is actionable: recentring the decision statistic on the empirical null lowers the certified minimum detectable effect roughly tenfold—from 0.15 to 0.02 at $q=2$ —restores size control, and yields the protocol’s first *admissible* certificate, for a bounded and now adequately powered exclusion of positive short-horizon serial dependence. Casting each gate as an e-value makes admission anytime-valid: one product certificate bounds the probability of ever falsely admitting a detector, under optional stopping, with no error budget across gates or monitoring times. A three-family exhibit (rolling-quantile, hidden-Markov, variance-ratio) yields cost-dominated, instrument-failed, and target-mismatched dispositions—failure reasons no performance comparison reports. Every value traces to frozen, version-pinned artifacts, with confirmatory and exploratory provenance separated per table cell.

Keywords: regime detection, model validation, severe testing, statistical power, equivalence testing, market microstructure, mutual information, backtesting

1. Introduction

Financial econometrics has a mature answer to the question “is this risk forecast usable?”: backtest it. Value-at-risk models must demonstrate correct unconditional and conditional coverage before their outputs enter capital calculations [1, 2], and supervisory guidance makes pre-use validation of quantitative models an explicit governance obligation [3]. The discipline exists elsewhere too: sequential changepoint detectors

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are *designed* to declared operating characteristics—average run length to false alarm against detection delay [4, 5]—and Markov-switching models have dedicated specification tests [6]. What is missing is the analogous step for regime-detector *outputs used as evidence*. A detector—a variance-ratio cascade, a Markov-switching filter, a volatility-quantile rule, a machine-learned classifier—converts a noisy market history into labels, triggers, and statistics, and those outputs are then used directly: as conditioning variables in asset-pricing regressions, as state inputs to allocation and risk systems, as the headline evidence that a market “switched regimes.” Size and power simulations are standard for test statistics studied in isolation [7], but they are not bound to the use. Four questions are rarely answered as a package, for a declared claim, before the output is used: whether the detector could have detected the effect size being claimed, how often it fires on realistic noise for *this* asset and frequency, whether its labels survive a change of bar-construction scheme, and whether its output carries target-relevant information net of costs.

This paper proposes and demonstrates a pre-use validation protocol for detector outputs, which we call *detector admissibility certification*. The unit of analysis is not the detector but a detector-output *claim*: a frozen detector, a declared target, a sampling scheme, a horizon, a signal family, an empirical null family, an equivalence margin, and a cost model, all fixed before evidence is interpreted. The protocol measures five operating characteristics for that claim—power, empirical size, transport across sampling schemes, detector-output information, and net information value after costs—and returns a disposition: the output is admitted as evidence for the declared claim, or admission is withheld for a stated, gate-level reason. The output of a certification run is an auditable certificate, not a score.

The philosophical basis is severity [8, 9]: a detector output supports a claim only to the extent that the detector was demonstrably capable of contradicting that claim and did not. Certification operationalizes severity for detector outputs. Power is demonstrated by controlled signal injection into the empirical data path, not assumed from asymptotic theory; false-alarm behavior is measured under dependence-preserving resampling, not read off a nominal level; and a claim made under one sampling scheme is not transported to another until equivalence has been certified within a preregistered margin. The protocol’s refusal states are as informative as its admissions: a withheld admission names the failed gate, which converts a generic null result into a measured property of the instrument [10].

We demonstrate the protocol by certifying a Holm-corrected Lo–MacKinlay variance-ratio cascade [11, 12] on BTCUSDT one-minute perpetual futures over 2021–2025 ($n = 2,415,148$ bars)—a market whose weak-form efficiency is actively contested [13, 14], which makes it a natural stress environment for detector claims. The worked detector is an example instrument, not the contribution. The certification findings are stated in bounded, minimum-detectable-effect form throughout: the preregistered injection grid excludes detection of variance-ratio departures $\delta \leq 0.10$ at all twelve window–horizon combinations; disclosed post-freeze positive controls bracket the response threshold in $(0.10, 0.15]$ at $q=2$ and $(0.20, 0.30]$ at $q=5$; the cascade’s nominal-level rule is strongly conservative under a circular-block empirical null (zero false alarms in 100 replicates, with a mis-centered asymptotic reference); transport across calendar, volume, and event-time bars is certified at $q=2$ and fails at $q=5$; and trigger information about forward return signs is statistically resolvable but its net value is convention-dependent, flipping sign between per-epoch and per-trigger cost attribution. A companion exhibit runs three detector families through the same gates and obtains three different dispositions for three different reasons, none of which is relative performance.

The paper makes three contributions.

1. *Scientific*. It binds existing validation disciplines into a certification operator for regime-detector outputs: a claim tuple frozen before interpretation, a characterization vector of measured operating characteristics, and a disposition taxonomy generated by an explicit precedence rule. This imports the validate-before-use posture of risk-model backtesting [1, 2, 3] and the design-to-operating-characteristics posture of sequential detection [4, 5] into a setting neither covers—detector outputs used as evidence—on the severity rationale of Mayo [8].
2. *Methodological*. It assembles the certification instruments and their decision semantics: injection-based power surfaces with exact finite-draw uncertainty and minimum-detectable-effect reporting; empirical-size measurement under dependence-preserving nulls; equivalence-tested transport across bar-construction schemes [15]; and a detector-output information gate whose economic reading requires

a declared cost-attribution convention. Individually these components are standard; the contribution is their binding to one frozen claim with a precedence rule and refusal states, which Section 2 argues no existing framework supplies.

3. *Empirical and reproducibility.* It exhibits a complete worked certificate for a regime detector on high-frequency data, with findings that surface only when the components share one frozen claim: a preregistered 96-cell power grid uniformly silent below $\delta = 0.10$; a horizon-discriminating cross-clock verdict (two of three scheme pairs certified at $q=2$, none at $q=5$); and a value gate whose sign depends on the cost convention—a dependence the certificate exposes rather than averages away. Every reported value traces to a frozen, version-pinned artifact via a claim-traceability ledger, with confirmatory and exploratory provenance distinguished at the level of individual table cells.

A detector output is admitted as evidence only after its operating characteristics are measured for the declared claim

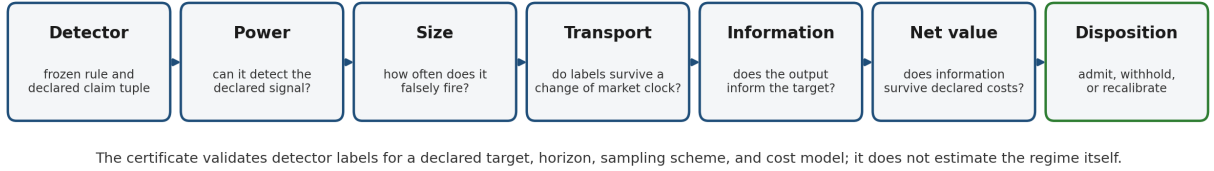


Figure 1: Certification pipeline. A detector output is admitted as evidence only after power, size, transport, information, and net-value gates have been measured for the frozen claim tuple. The certificate validates detector labels for a declared target, horizon, sampling scheme, and cost model; it does not estimate the regime itself.

Section 2 positions certification against the validation, severity, data-snooping, market-clock, and information-value literatures. Section 3 defines the claim tuple, the gates, the certificate, and the disposition rules. Section 4 certifies the variance-ratio cascade on BTCUSD. Section 5 runs three detector families through the same protocol. Section 6 develops what certification adds relative to comparison testing, and Section 7 states the domain of validity and the sequential extension. Appendices contain derivations, the ETHUSD replication, the EUR/USD cost instantiation, and the reproducibility package.

2. Positioning and Related Work

2.1. Validation before use

The nearest methodological ancestor of certification is the backtesting of risk forecasts. Kupiec [1] tests whether a value-at-risk model’s violation frequency matches its nominal coverage; Christoffersen [2] adds conditional coverage. Both are pre-use gates: a model whose violations are mis-calibrated is not trusted, whatever its in-sample fit. Supervisory model-risk guidance generalizes the posture—models require validation commensurate with their use, ongoing monitoring, and explicit use restrictions [3]. Certification transfers this discipline to regime detectors, whose outputs are neither forecasts nor violation streams and therefore fall outside existing backtesting practice. Three gates have no analogue in the VaR lineage and are the substance of the transfer: power against a declared signal family, transport across sampling schemes, and information value net of costs.

2.2. Severity and the value of instrumented negatives

The certificate’s logic is the severity requirement of Mayo [8] and Mayo and Spanos [9]: evidence counts only when produced by a procedure with a demonstrated capacity to have found the claim false. In our setting the demonstration is literal—known-amplitude signals are injected into the empirical path and the detector’s recovery probability is measured. The inferential payoff is that non-detection becomes quantitative: “the detector was silent” becomes “the detector was silent, and it detects $\delta = 0.15$ departures with probability

one at this horizon, so amplitudes above 0.15 are excluded.” This is the minimum-detectable-effect reporting format standard in experimental sciences, and it is what makes negative results publishable evidence rather than absence of evidence [10, 16]. Signal injection as an instrument-calibration device has a developed engineering practice in gravitational-wave detection [17]; we import the practice, not the physics.

2.3. What certification is not—and the gaps it closes

A large literature controls data-snooping when many rules are compared on one sample: the reality check [18], stepwise extensions [19], superior predictive ability [20], the model confidence set [21], and their applications to trading rules [22]. Multiple-testing discipline for anomaly discovery makes the same point at the scale of the literature itself [23, 24]. These methods answer a *relative* question—does the best of a universe beat a benchmark, and which candidates are statistically indistinguishable from the best—and they presuppose that each candidate’s output is a valid measurement of a common object. Certification answers the prior, *absolute* question for a single output and a single claim: is this label stream a valid measurement at all, for this target, at this horizon, under this sampling scheme and cost model? The two are complements, not substitutes: a detector can enter a model confidence set while its labels fail transport certification, and a certification pass says nothing about relative performance (Section 6 develops the contrast on our own artifacts).

No component check below is new, and the paper claims no novelty for power curves, resampled nulls, equivalence tests, or information bounds. Run separately, however, the components leave three gaps. First, there is no single pre-declared object that ties detector, target, sampling scheme, signal family, null family, cost model, and horizon together before evidence is interpreted, so each check can be (and in practice is) run under silently different conditions. Second, there is no precedence rule for multiple failures—for example, a sensitive detector that is transport-uncertified and cost-dominated—so ad hoc narratives fill the gap. Third, there is no auditable output that tells a downstream user whether the detector may speak for the claim at hand, must abstain, or must be recalibrated. The certificate closes all three, and the closure has bite: several of our findings (the uniform silence of the preregistered power grid, the horizon-discriminating transport verdict, the convention-dependence of the value gate) are visible only when the components are forced to share one frozen claim.

2.4. Sampling schemes and market clocks

That the sampling scheme is part of the measurement, not a nuisance choice, is an old point in market microstructure: transaction, volume, and subordinated clocks change the distributional properties of returns [25, 26, 27, 28, 29, 30]. A detector calibrated on calendar bars is therefore not automatically portable to volume or event-time bars. Certification treats portability as a statistical hypothesis to be certified—practical equivalence of a declared output statistic within a preregistered margin [15], with multiplicity control [12]—rather than as an assumption. We use “transport” in the spirit of the transportability literature, which asks when results carry from one data-generating environment to another [31]; here the environments are bar-construction schemes over the same market history. An event-time (intrinsic) clock advances when a pre-declared market movement or activity event occurs; horizons are aligned across schemes by a declared calibration table before any comparison is made.

2.5. Information value and its ceiling

The economic reading of detector output uses the growth-optimal bound: side information about a payoff-relevant target raises attainable log-growth by at most the mutual information between signal and target [32, 33, 34, 35]. Two disciplines follow. First, the relevant estimand is the information in the *detector output*, $I(D; Y)$, not the information in the raw path: by data processing, a causal output cannot exceed the history it compresses, so raw-path information is a scale diagnostic, never detector evidence. Second, an upper bound net of costs is a use gate, not a performance claim: the volatility-timing literature shows that economic value claims live or die on cost and implementation assumptions [36, 37], and the certificate makes the cost convention an explicit, frozen field.

2.6. Abstention, calibration, and certificates

Certification’s refusal states have counterparts in selective prediction and classification-with-rejection, where a model abstains when its output is not trustworthy [38], in conformal prediction’s validity guarantees [39], and in the calibration literature’s insistence that confidence match frequency [40]. Machine-learning governance has converged on structured pre-use disclosure documents—model cards reporting a model’s evaluation conditions and intended-use boundaries [41]—and measurement science supplies the reporting form itself: an instrument is accompanied by a calibration certificate stating what was measured, under what conditions, with what uncertainty, and with what validity limits [42]. The detector certificate is that form, instantiated for statistical instruments, with the difference that its fields are measured gates with decision consequences rather than descriptive disclosures.

3. The Certification Protocol

3.1. Claim tuple and operator

Let a detector-output claim be

$$\mathcal{A} = (D_\theta, Y, g, W, q, s, \mathbb{P}_0^*, \epsilon_g, Q, C_{\text{rt}}),$$

where D_θ is a frozen detector, Y is the forward scientific or economic target, g is the sampling scheme (market clock), (W, q) defines the detector window and horizon, s is the declared signal family, $\mathbb{P}_0^*$ is the empirical null family, ϵ_g is the transport-equivalence margin, and (Q, C_{rt}) define the decision size and cost model. Certification is the operator

$$\mathcal{O}(\mathcal{A}) = (C_D, \text{disposition, failure profile}), \quad C_D = \{P_{\text{det}}, \alpha_D^*, \Delta_D, I_D, W_D\},$$

mapping the claim to a measured characterization vector—power (P_{det}), empirical size (α_D^*), transport discrepancy (Δ_D), detector-output information (I_D), and net information value (W_D)—plus a disposition and a gate-level failure profile. Throughout, “admissible” is used in the evidentiary sense—whether an output may be used as evidence for the declared claim—not in the decision-theoretic sense of Wald; and “net information value” is a cost-adjusted Kelly-growth bound, not the expected value of information of decision analysis. The detector family is unrestricted: classical test statistics, state-space filters, volatility rules, or learned classifiers are all admissible arguments, because the operator evaluates outputs, not internals.

The operator runs seven stages on the frozen tuple: *freeze* (fix all tuple fields and record the commit), then the five measurement gates *power* P_{det} , *size* α_D^* , *transport* Δ_D , *information* I_D , and *net value* W_D defined below, and finally a *disposition* with its failure profile. Figure 1 shows the ordering.

3.2. Certificate, governance, and precedence

A certification run produces a certificate rather than a scalar score; Table 1 lists the minimum fields. The claim owner—study author or preregistration owner in research; model owner plus validation, risk, and execution stakeholders in production—must freeze the tuple before the output is used. Any change in detector code, target, bar construction, venue, fee schedule, or market-structure condition triggers recertification or an explicit monitoring exception.

Disposition precedence is conservative and fixed in advance by the following rule, applied mechanically:

1. *Instrument validity*: if frozen validity gates fail, the disposition is *instrument-failed* and no statistical gate is interpreted.
2. *Target eligibility*: if the detector measures a different object than the claim target, the disposition is *target-mismatched*.
3. *Size*: if empirical false-alarm behavior is inconsistent with the declared level (in either direction), the disposition is *size-distorted*.

Table 1: Minimum certificate fields.

Field	Required content
Claim tuple	Frozen detector, target Y , sampling scheme g , horizon (W, q) , signal family s , null family, equivalence margin, decision size, cost model, and cost-attribution convention.
Power	Executed grid, draws per cell, fire rates, exact intervals, and response thresholds or explicit brackets; confirmatory versus exploratory provenance per cell.
Size	Null families, block or resampling parameters, number of draws B , Monte Carlo resolution at B , empirical false-alarm estimate, and the calibration of the nominal reference.
Transport	Statistics Φ , scheme pairs, margin rationale, multiplicity adjustment, per-pair certification status, and precision limits of non-certified rows.
Information/value	Detector output used, target, estimator, uncertainty and permutation reference, and net value under the declared convention (with sensitivity to the attribution convention reported).
Failure profile	Gate-level reasons for non-admission, in precedence order, with all failed gates retained.
Governance	Certificate owner, artifact hashes or paths, freeze commit, expiry conditions, recalibration triggers, and downstream action.

4. *Transport*: for transported claims, if required equivalence is not certified, the disposition is *transport-uncertified*.
5. *Power*: if P_{det} does not reach the required level on the declared amplitude family, the disposition is *underpowered*.
6. *Information/value*: if I_D is statistically unsupported or $W_D \leq 0$ under the declared convention, the disposition is *cost-dominated*.

Missing required estimands yield *incomplete* unless the available evidence already fails a gate above, in which case that gate’s disposition is reported and the missing coverage is recorded in the failure profile. When several gates fail, the certificate reports the first failure in this order and retains the rest in the profile. Every disposition in this paper, including the exhibits of Sections 4 and 5, is generated by this rule. “Admissible” always means admissible *for this claim tuple*—never profitable, never generally valid.

3.3. Objects and gates

Let $X = (r_1, \dots, r_T)$ be an empirical return path. A sampling scheme $g \in \mathcal{G}$ maps the same market history into a scheme-specific bar path $T_g(X)$; in the demonstration $\mathcal{G}$ contains calendar, volume, and event-time bars. A frozen detector is a causal map $D_\theta(T_g(X)) \mapsto (F_t^g, L_t^g, S_t^g, M_t^g)$: trigger stream, label stream, output statistic, and validity metadata, with θ frozen before characterization. For a signal family s and amplitude δ , an injection operator $J_{\delta,s}$ produces a perturbed empirical path; the detector runs unchanged on the transformed, injected path.

Power.. The central estimand is

$$P_{\text{det}}(\delta; W, q, g, s) = \Pr\{D_\theta(T_g(J_{\delta,s}(X))) \text{ fires}\}, \quad (1)$$

with response thresholds $\delta_p = \inf\{\delta : P_{\text{det}}(\delta; \cdot) \geq p\}$ for declared p (here $p \in \{0.50, 0.90\}$). We name δ_{90} the *certified minimum detectable effect* (cMDE): the smallest declared-family amplitude the frozen detector recovers with probability at least 0.90, on this asset, frequency, horizon, and sampling scheme. The cMDE is intended as a portable reporting primitive—the power analogue of a value-at-risk exception rate—so that “detector D has cMDE = 0.15 at ($q=2$, 1-min BTCUSDT)” becomes a comparable, citable statement of instrument sensitivity, and a negative result acquires a severity: departures above the cMDE are excluded with demonstrated power, departures below it are not claimed either way. When the executed grid does not identify δ_{90} , the certificate reports a bracket, never an extrapolation, and a cMDE from post-freeze amplitudes is labeled exploratory-grade until a preregistered confirmatory rerun.

Proposition 1 (Power-surface estimation). *For a fixed detector rule, scheme, injection family, and grid cell, Monte Carlo injection outcomes are Bernoulli draws with success probability P_{det} . The sample fire rate estimates the response surface conditional on the empirical path and injection design, and exact binomial intervals give the finite-draw uncertainty.*

The proposition is deliberately conditional—it characterizes the frozen instrument on the declared noise path and signal family, and claims no unconditional market law. Proofs of this and the following statements are immediate and collected in Appendix A.

Size.. Asymptotic references can fail in high-frequency data with heavy tails, volatility clustering, and overlapping-window aggregation. Certification therefore measures

$$\alpha_D^*(W, q, g) = \Pr_{\mathbb{P}_0^*} \{D_\theta(T_g(X^*)) \text{ fires}\}, \quad (2)$$

under dependence-preserving nulls (circular block, stationary block, wild, or sign-symmetric resampling; 43, 44). The certificate must declare the draw budget B and report the exact binomial resolution at that budget; a low- B diagnostic can demonstrate the gate but cannot certify size for deployment. Size distortion in either direction is reportable: an over-sized rule invalidates positives, and a strongly conservative rule silently destroys power—a fact the power gate will independently register.

Transport.. For a declared output statistic Φ and scheme pair (g, h) , the transport discrepancy is

$$\Delta_D(g, h; \Phi) = |\Phi(D_\theta(T_g(X))) - \Phi(D_\theta(T_h(X)))|, \quad (3)$$

and portability is certified only if two one-sided equivalence tests place Δ_D inside the preregistered margin ϵ_g after multiplicity control across the declared family of pairs and horizons [15, 12].

Proposition 2 (Transport non-certification). *If practical equivalence is not certified for a pre-declared statistic Φ , then detector claims calibrated under scheme g are not certified under scheme h without recalibration or a scheme-specific certificate.*

Non-certification is not a demonstration of difference: a pair can fail because the estimated discrepancy exceeds the margin or because the comparison lacks precision (event-time bars are far scarcer than calendar bars). The certificate distinguishes the two by reporting the interval against the margin.

Information and net value.. Let Y_{t+q} be the declared target and D_t^g the detector output available before the decision. The gate estimand is the detector-output information

$$I_D(q, g) = I(D_t^g; Y_{t+q}), \quad (4)$$

never a raw-path quantity such as $I(\text{sign } r_t; \text{sign } r_{t+q})$. For decision size Q and all-in round-trip cost $C_{\text{rt}}(q, Q, g)$, the net information value in basis points per decision epoch is

$$W_D(q, Q, g) = 10,000 I_D(q, g) - C_{\text{rt}}(q, Q, g) \cdot \rho, \quad (5)$$

where I_D is in nats under the log-optimal fair-odds bound [32, 33, 35] and ρ encodes the declared cost-attribution convention: under *per-epoch* attribution every decision epoch bears the full round-trip cost ($\rho = 1$); under *per-trigger* attribution cost is borne only when the detector’s action state is on, so ρ equals the trigger rate. The conversion is an idealized gross upper bound on log-growth, not a realized-profit forecast, and the cost term must include commissions, spread, slippage, impact, funding, latency, and non-fill risk where relevant. Equation (5) makes the convention a certificate field because, as Section 4 shows, the sign of W_D can depend on it.

Proposition 3 (Cost gate). *If $W_D(q, Q, g) \leq 0$ under the declared convention, detector-only use is excluded for the stated target and cost model. $W_D > 0$ is necessary but not sufficient for economic use: it is an upper bound on attainable growth, not an attainable strategy.*

Corollary 1 (Data-processing ceiling). *If D_t^g is a causal function of observed history, $I(D_t^g; Y_{t+q}) \leq I(X_{\leq t}^g; Y_{t+q})$: a detector output cannot be credited with information that is only present in the raw history it compresses.*

3.4. Certificate-level error control via e-values

The gates that are hypothesis tests—size, transport equivalence, and information—admit a common error currency that the power and value estimation gates do not require: the e-value. An e-value for a null H_0 is a nonnegative statistic E with $\mathbb{E}_{H_0}[E] \leq 1$; by Markov’s inequality $\Pr_{H_0}(E \geq 1/\alpha) \leq \alpha$, so certifying a gate at level α means observing $E \geq 1/\alpha$ [45, 46]. Writing each testing gate k as an e-value E_k for its inadmissibility null H_0^k turns the certificate’s conjunction of gates into a single error-controlled object.

Proposition 4 (Static admission control, no multiplicity penalty). *Suppose admission requires certifying every required testing gate, $E_k \geq 1/\alpha$ for all $k \in K$. Under the global inadmissibility null $H_0 = \bigcup_{k \in K} H_0^k$ (the detector violates at least one gate), $\Pr_{H_0}(\text{admit}) \leq \alpha$, with no correction for the number of gates.*

Proposition 5 (Anytime-valid admission control). *Suppose each gate is monitored by an e-process $(E_t^k)_{t \geq 0}$ —a nonnegative supermartingale under H_0^k with $\mathbb{E}[E_0^k] \leq 1$ —and admission at time t requires $E_t^k \geq 1/\alpha$ for all k . Then over an unbounded monitoring horizon and any data-dependent stopping rule, $\Pr_{H_0}(\exists t : \text{admit}) \leq \alpha$.*

Proofs are in Appendix A. Proposition 4 is the e-value form of the intersection–union test [47], of which the transport TOST gate is the textbook case; the free multiplicity follows because admission is a conjunction, so the worst case is governed by the single hardest gate. Proposition 5 is Ville’s inequality applied to the failing gate’s e-process. Together they replace the fixed error budget split across gates and the per-window maximum-statistic multiplicity of a p-value monitor with one bound that holds continuously, under optional stopping—the property a repeatedly consulted p-value monitor lacks [48]. For downstream propagation the certificate reports the product $\prod_k E_k$ when gate e-values are built on separate data folds and the average $|K|^{-1} \sum_k E_k$ otherwise, the latter an e-value under arbitrary dependence.

The information gate instantiates this empirically. Its permutation p-values calibrate to e-values through the a-priori calibrator $f_\kappa(p) = \kappa p^{\kappa-1}$ at $\kappa = \frac{1}{2}$, giving $E = 2.17$ at $q=2$ ($p = 0.053$) and $E = 15.8$ at $q=5$ ($p = 0.001$); both fall short of the $1/\alpha = 20$ admission threshold at $\alpha = 0.05$, the conservative price of a distribution-free calibration. The deployment form is tighter and unifies two gates: a sequential bettor staking the detector’s forward-sign prediction accumulates wealth that is an e-process for independence, and whose log-growth rate is exactly the Kelly quantity of the value gate W_D —so the information gate’s evidence and its economic value are one object viewed two ways.

3.5. Run integrity

Certification is only as credible as its bookkeeping, so the protocol imports reproducibility controls as inferential requirements, not engineering conveniences. The claim tuple is frozen in a version-controlled preregistration commit; every artifact records that commit, its own code commit, and its random seeds; evidence produced after the freeze must carry an explicit exploratory flag in its provenance block, and the certificate must disclose that flag wherever the evidence is displayed; data outside the declared development span is excluded by the loader, and each artifact records the exclusion flag. Each of these controls is engaged—and its consequences disclosed—by the demonstration in Section 4.

4. Certifying a Variance-Ratio Cascade on BTCUSDT

4.1. Data, detector, and declared claim

The demonstration uses BTCUSDT one-minute perpetual futures over 2021–2025 ($n = 2,415,148$ bars; effective span 2021-05-29 to 2025-12-31). The market is treated as an empirical noise environment for instrument characterization, not as the object of an efficiency claim. The worked detector is a Holm-corrected Lo–MacKinlay variance-ratio cascade [11, 7, 12]: for horizon q , the variance-ratio departure is computed from overlapping q -period returns inside non-overlapping windows of width W ; the cascade fires when at least one horizon in $\{2, 5, 15, 60\}$ has a Holm-adjusted two-sided p-value at or below 0.05 and a positive median standardized statistic. The positive sign gate makes the instrument a positive-autocorrelation trigger, not a general $|\text{VR} - 1|$ detector. Multiple-horizon variance-ratio testing otherwise follows the joint-testing lineage

of Chow and Denning [49], with finite-sample refinements available through rank/sign and wild-bootstrap variants [50, 51].

The declared claim to be certified is: *the cascade’s trigger stream may be used as evidence of positive short-horizon serial dependence on calendar-time bars—primary horizon $q=5$, window $W=120$ —and its triggers may be read economically under a 10bps all-in round-trip cost convention at unit decision size.* The signal family s is a positive AR(1) injection calibrated to target $|\text{VR}(q) - 1|$ amplitudes; the null family is a centered circular block bootstrap; the transport margin is the preregistered minimum detectable effect, $\epsilon_g = \text{MDE}(q, W, n_{\text{eff}}; \alpha=0.05, 1-\beta=0.80) = 0.019748$ on the median $|\text{VR} - 1|$ scale, which prose elsewhere rounds to 0.02. The tuple was frozen in preregistration commit `1dc5c82` (v4.0, 2026-06-11), with an earlier v3.0 freeze lineage (2026-06-05) governing the holdout and replication runs. The bootstrap block length 120 and the Holm family size 12 are implementation choices made after the freeze, which fixed the resampling rule and the correction method but not those two integers; both are recorded in `prereg/DEVIATIONS.md` (D3, D5). Artifacts pin these commits and seed 43, except the persistence null, which records seed 42 in its own artifact. The frozen protocol documents and their machine-readable commit and run timestamps are archived at `10.6084/m9.figshare.33106934`. That deposit fixes document content only: git committer timestamps are self-reported, and the deposit post-dates the cited runs, so neither constitutes third-party attestation of the freeze-before-run ordering.

4.2. Power: a preregistered exclusion grid and disclosed positive controls

The preregistered power design is a 96-cell grid: amplitudes $\delta \in \{0.0005, 0.001, 0.002, 0.005, 0.01, 0.02, 0.05, 0.10\}$, horizons $q \in \{2, 5, 15, 60\}$, windows $W \in \{60, 120, 240\}$, with 200 Monte Carlo injection draws per cell. The confirmatory result is uniform: *in all 96 preregistered cells the cascade fired on 0 of 200 draws* ($\hat{P}_{\text{det}} = 0$, exact 95% interval $[0, 0.0183]$ per cell). In minimum-detectable-effect form: for this detector class, injection design, and noise path, detection of variance-ratio departures $\delta \leq 0.10$ is excluded at every tested window–horizon combination, with per-cell fire probability bounded above by 1.8% at 95% confidence.

After the freeze, eight additional cells were run at larger amplitudes as positive controls. Their provenance blocks carry an explicit exploratory flag, and we report them as exploratory evidence, separately from the confirmatory grid (Table 2). They establish that the harness detects large signals— $\delta = 0.15$ at $q=2$ fires on every draw at all three windows—and they bracket the response thresholds: $\delta_{90}(q=2) \in (0.10, 0.15]$ for all tested W ; $\delta_{90}(q=5, W=120) \in (0.20, 0.30]$; at $q=5$ with $W \in \{60, 240\}$ the threshold exceeds 0.15 and is unbracketed above on the executed grid; at $q \in \{15, 60\}$ it exceeds 0.10 and is unbracketed. Because the bracketing amplitudes were executed post-freeze, these brackets are exploratory-grade: a certificate-grade threshold statement requires a post-registered confirmatory rerun at the bracketing amplitudes. The design lesson is recorded rather than hidden: a confirmatory amplitude grid should be frozen only after a pilot power scan brackets the response region, which the v4.0 design did not do—the addendum exists because of that gap, and the disclosure obligation is the protocol registering its own design debt. Under the declared claim—whose scientifically interesting amplitudes lie at or below 0.10—the instrument is *underpowered*: the observed market departure at the primary cell (median $|\text{VR}(5) - 1| = 0.1603$) sits between the excluded region and the $q=5$ response threshold, in a range where the cascade has never demonstrated recovery.

The sharpness of the transition—0/200 at one executed amplitude, 200/200 at the next—is a property of the cascade’s aggregated decision statistic discussed next, and the size gate explains why the thresholds are as high as they are.

4.3. Size: empirical false-alarm behavior and reference calibration

The size gate was audited at the two cascade-relevant cells ($W=120, q \in \{2, 5\}$) under a centered circular block bootstrap of the full return path (block length 120, $B = 100$, seed 43). Two findings.

First, the false-alarm estimate: at nominal level 0.05, the asymptotic two-sided rule fired on 0 of 100 null replicates in both audited cells ($\hat{\alpha}_D^* = 0$, exact 95% interval $[0, 0.0362]$). Since the interval excludes the nominal 0.05, the empirical level is significantly below nominal: by gate 3 of the precedence rule (Section 3) this is a certified *size distortion in the conservative direction*, not a pass. Two scope caveats are part of the finding. The audit covers the two cascade-relevant per-horizon cells, not the four-horizon union rule;

Table 2: Power gate. Confirmatory grid (preregistered, freeze 1dc5c82) and exploratory positive controls (post-freeze, `exploratory_addendum: true` in artifact provenance). All cells use 200 draws; intervals are exact Clopper–Pearson 95%.

Provenance	Cells	Fires / draws	$\widehat{P}_{\text{det}}$	95% interval
Confirmatory	all 96: $\delta \leq 0.10$, $q \in \{2, 5, 15, 60\}$, $W \in \{60, 120, 240\}$	0 / 200 each	0.00	[0.0000, 0.0183]
Exploratory	$\delta = 0.15$, $q = 2$, $W \in \{60, 120, 240\}$	200 / 200 each	1.00	[0.9817, 1.0000]
Exploratory	$\delta = 0.15$, $q = 5$, $W \in \{60, 120, 240\}$	0 / 200 each	0.00	[0.0000, 0.0183]
Exploratory	$\delta = 0.20$, $q = 5$, $W = 120$	0 / 200	0.00	[0.0000, 0.0183]
Exploratory	$\delta = 0.30$, $q = 5$, $W = 120$	200 / 200	1.00	[0.9817, 1.0000]

Response-threshold brackets implied by the executed grid: $\delta_{90}(q=2, \cdot) \in (0.10, 0.15]$; $\delta_{90}(q=5, W=120) \in (0.20, 0.30]$; $\delta_{90}(q=5, W \in \{60, 240\}) > 0.15$; $\delta_{90}(q \in \{15, 60\}, \cdot) > 0.10$. No interpolation is performed.

Intervals are exact Clopper–Pearson.

The $\delta = 0.15$, $q=5$, $W=240$ cell is a declared W -invariant clone of the $W=120$ cell at the same amplitude and horizon (the cascade uses $W_{\text{primary}}=120$ there), not an independent Monte Carlo run. It is the only clone among the 104 cells; the 96 confirmatory cells are all independent runs.

The exploratory addendum registered 48 cells ($\delta \in \{0.15, 0.20, 0.30, 0.50\}$ crossed with the q and W grids); 8 were executed. The unexecuted cells sit at the higher amplitudes, where detection is more likely, so the executed subset is conservative with respect to the exclusion claim and uninformative about the upper brackets.

Five of the twelve (W, q) combinations fall below the Lo–MacKinlay stability guideline $W/q \geq 10$ —(60, 60), (120, 60), (240, 60), (60, 15) and (120, 15)—so those rows sit outside the regime in which the variance-ratio sampling theory is well behaved, and their reading is correspondingly weaker. All are silent, so the exclusion conclusion does not turn on them. See `prereg/DEVIATIONS.md D2`.

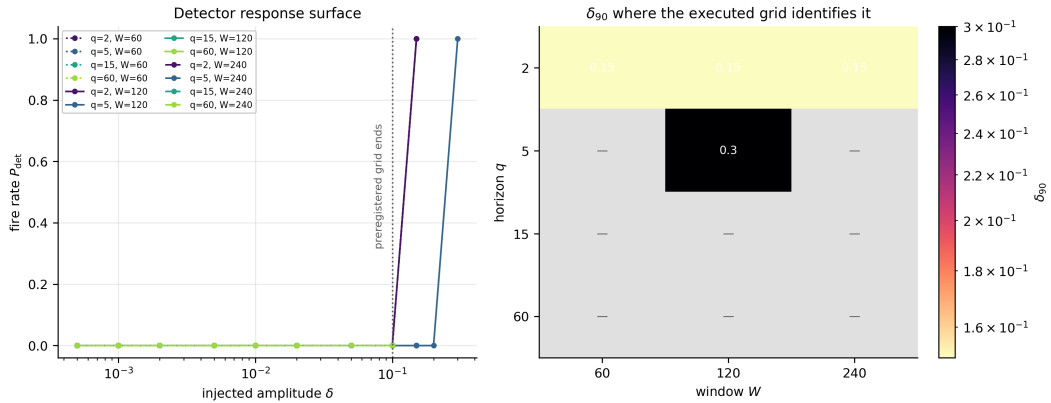


Figure 2: Power surface for the frozen cascade. Left: fire rate against injected amplitude for all twelve (q, W) combinations; every preregistered cell ($\delta \leq 0.10$, dotted vertical line) is at zero, and the rise to certain detection occurs only in the exploratory region ($q=2$ at $\delta = 0.15$; $q=5$, $W=120$ at $\delta = 0.30$). Right: implied δ_{90} where the executed grid identifies it; grey cells are unbracketed above. The step from 0/200 to 200/200 between adjacent executed amplitudes reflects the grid resolution, not a measured discontinuity.

the full-cascade false-alarm rate on the same 100 resampled paths is a required extension for a deployment certificate. And the executed null is a single dependence-preserving family; the wild-bootstrap null, which imposes the martingale-difference hypothesis the variance-ratio statistic actually tests [51], is part of the declared certificate-grade family and was not run at this diagnostic budget.

Second, the reference calibration: the cascade aggregates window-level standardized statistics into a median, and the artifact records where that median actually lives under the dependence-preserving null—centered at -0.165 ($q=2$) and -0.389 ($q=5$), with 5th–95th percentile spans of only 0.029 and 0.023 respectively. The $N(0,1)$ asymptotic reference used by the frozen rule is therefore mis-centered and mis-scaled for this aggregated statistic: the observed market medians (-0.1819 at $q=2$, -0.4017 at $q=5$) receive asymptotic two-sided p-values of 0.856 and 0.688, while their positions in the empirical null distribution give 0.040 and 0.050 (both below the null 5th percentile; Holm-adjusted over the two audited cells, 0.079). Neither ingredient of this miscalibration is mysterious after the fact: the negative center is the known finite-sample bias of overlapping variance-ratio statistics [7], and the tight dispersion is what aggregating $\sim 20,000$ window statistics into a median produces. The gate’s value is not discovery but registration: the frozen rule shipped with a foreseeable miscalibration unhandled, and the certificate records it against the rule rather than leaving it implicit in the p-value formula. The direction also matters: both observed medians lie on the *negative* side, so the one-sided positive gate correctly withholds (empirical positive-gate p-values 0.970 and 0.960), and no confirmatory market claim arises from these cells.

Table 3: Size gate at the audited cells ($W=120$; centered circular block bootstrap, block length 120, $B = 100$).

q	$\hat{\alpha}_D^*$ (95%)	Null median (p05, p50, p95)	Obs. median Z	Asympt. p	Empir. p	Pos.-gate p
2	0/100 [0, 0.036]	($-0.180, -0.165, -0.151$)	-0.1819	0.856	0.040	0.970
5	0/100 [0, 0.036]	($-0.401, -0.389, -0.378$)	-0.4017	0.688	0.050	0.960

$\hat{\alpha}_D^*$: share of null replicates on which the nominal-level asymptotic two-sided rule fired. “Empir. p”: two-sided position of the observed median in the null distribution of medians (Holm-adjusted over the two audited cells: 0.079 each). $n = 20,126$ non-overlapping windows per replicate. A deployment certificate would rerun this gate at the Monte Carlo budget declared in Table 1; $B = 100$ demonstrates the gate and bounds, but does not resolve, the false-alarm rate.

The size and power gates now explain each other, which is the point of binding them in one certificate: a rule whose reference is mis-centered by several null standard deviations in the conservative direction almost never rejects—so its false-alarm rate is near zero *and* its response thresholds are far above the amplitudes of scientific interest. The certificate records the pair as two consistent failures (size-distorted in the conservative direction; underpowered below $\delta = 0.15$), which is a diagnosis, not a defense, of the frozen instrument.

4.4. Transport: equivalence across calendar, volume, and event-time bars

The transport gate compares the median $|VR - 1|$ departure across calendar (clock), volume, and event-time (intrinsic) bar constructions of the same market history, with horizons aligned by a frozen calibration table (median bar durations: 1, 4, and 22 minutes respectively). Equivalence within $\epsilon_g = 0.019748$ is tested per scheme pair and horizon by two one-sided tests, Holm-adjusted across the full 12-comparison family. Table 4 reports every row; nothing is omitted.

The margin is not a round number by choice. Preregistration v4.0 §4.6 fixes it as the minimum detectable effect at $\alpha = 0.05$ and power 0.80 under the Pre-Check B convention, $MDE = (z_{1-\alpha/2} + z_{1-\beta})/\sqrt{n_{\text{eff}}}$, evaluated at the calendar gauge’s $n_{\text{eff}} = 20,126$ non-overlapping windows: 0.019748. One margin governs the whole family. Evaluating n_{eff} per pair instead would give the event-time pairs, which have only 454 windows, a margin of 0.132, under which every comparison certifies trivially; the single calendar-anchored margin is the conservative reading and is the one the original run implemented. The 1.3% gap between 0.019748 and its rounded form 0.02 is not cosmetic: it decides the $q=2$ clock-intrinsic row, which certifies under 0.02 (Holm 0.0497) and does not under the preregistered margin (Holm 0.0682). `prereg/DEVIATIONS.md` D1 records the full history.

The verdict is horizon-specific (Figure 3). At $q=2$ two of the three scheme pairs certify: short-horizon cascade labels are portable between calendar and volume bars and between volume and event-time bars, but calendar-to-event-time portability is not certified (Holm 0.0682), even though its interval lies inside the

Table 4: Transport gate, BTCUSDT 2021–2025: all 12 preregistered comparisons. Pre-registered margin $\epsilon_g = 0.019748$ on median $|VR - 1|$; 90% intervals; Holm over the 12-row family. “Certified” requires Holm-adjusted equivalence.

q	Scheme pair	$\hat{\Delta}_D$	90% CI	Holm p	Certified
2	clock–volume	0.0093	[0.0068, 0.0118]	5.7×10^{-11}	yes
2	clock–intrinsic	0.0142	[0.0105, 0.0179]	0.0682	no
2	volume–intrinsic	0.0049	[0.0007, 0.0090]	2.3×10^{-8}	yes
5	clock–volume	0.0231	[0.0179, 0.0283]	1.000	no
5	clock–intrinsic	0.0351	[0.0260, 0.0442]	1.000	no
5	volume–intrinsic	0.0120	[0.0023, 0.0217]	0.758	no
15	clock–volume	0.0237	[0.0163, 0.0312]	1.000	no
15	clock–intrinsic	0.0279	[0.0007, 0.0550]	1.000	no
15	volume–intrinsic	0.0041	[−0.0233, 0.0315]	1.000	no
60	clock–volume	0.0040	[−0.0079, 0.0159]	0.135	no
60	clock–intrinsic	0.0014	[−0.0278, 0.0306]	1.000	no
60	volume–intrinsic	−0.0026	[−0.0335, 0.0283]	1.000	no

Non-overlapping window counts at $q=5$, W -equivalent 120: clock 20,126; volume 3,500; intrinsic 454. Rows whose interval width exceeds the two-sided margin ($2\epsilon_g = 0.0395$; e.g., all intrinsic pairs at $q \geq 15$) are precision-limited: equivalence could not have been certified at any discrepancy, and non-certification there is an evidence statement, not a difference finding. At $q=5$ the clock–volume and clock–intrinsic point estimates exceed the margin itself. The $q=2$ clock–intrinsic row is the one the margin decides: its Holm-adjusted p is 0.0682 under the preregistered $\epsilon_g = 0.019748$ and 0.0497 under the rounded 0.02, so it fails to certify by a margin narrower than the rounding of the margin itself. Its 90% interval, [0.0105, 0.0179], does lie entirely inside ϵ_g ; the preregistration is explicit that interval containment is not an equivalence test and may not stand in for TOST, so the row is reported as not certified. A certificate consumer should read this row as precision-adjacent rather than as a demonstrated discrepancy.

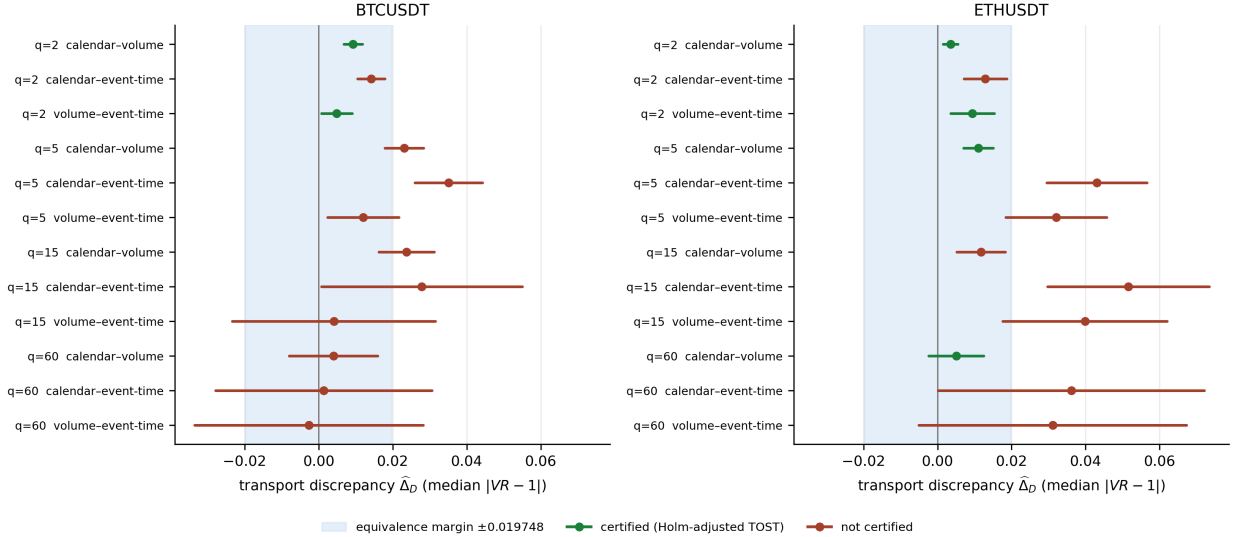


Figure 3: Transport gate as an equivalence forest plot: 90% intervals for the transport discrepancy against the preregistered ± 0.019748 margin (shaded), BTCUSDT (left) and ETHUSDT (right). Certification requires Holm-adjusted equivalence across the 12-row family, not merely an interval inside the band—hence rows such as BTC $q=2$ clock–intrinsic (interval inside the band, Holm $p = 0.0682$), ETH $q=2$ clock–intrinsic (Holm $p = 0.154$) and BTC $q=60$ clock–volume (Holm $p = 0.135$) remain uncertified: the multiplicity discipline binds.

margin. The pair that fails is the one with the fewest effective windows, which is what one would expect if the obstacle is precision rather than a real discrepancy—but the gate does not award certification for a plausible explanation. At the primary horizon $q=5$ no pair certifies, and the two clock-anchored discrepancies exceed the margin outright—volume and event-time bars attenuate the calendar-time departure (medians $0.160 \rightarrow 0.137 \rightarrow 0.125$). At $q \geq 15$ the event-time comparisons are precision-limited (454 windows). Under Proposition 2, the declared $q=5$ claim is *transport-uncertified*: calendar-bar labels are not certified evidence about volume- or event-bar regimes at that horizon, while the same statement at $q=2$ is certified. The gate passes where bar construction is immaterial and fails where it is material; the ETHUSD replication (Appendix B) repeats the pattern with a different certified set, confirming that transport certification is an asset- and horizon-specific measurement rather than a fixed property of the detector.

4.5. Information and net value: the convention is load-bearing

The information gate estimates $I_D = I(D_t; \text{sign } r_{t+q})$ for the cascade’s binary trigger on non-overlapping $W=120$ windows (20,126 labels), using the plug-in estimator on the 2×2 label–target table, with bootstrap intervals and a permutation reference (Table 5). Two estimation caveats bound what these numbers can support. The plug-in estimator’s first-order (Miller–Madow) bias for a 2×2 table is $1/(2n) \approx 2.5 \times 10^{-5}$ nats upward—about 0.25 bps, i.e., a fifth of the $q=2$ estimate—so small positive I_D values should be read net of that order. And the permutation reference shuffles label–target pairings, which imposes exchangeability at the window scale; a block-permutation reference consistent with the certificate’s own dependence-preservation requirement is the certificate-grade instrument and was not run at this budget. With those caveats, the triggers are sparse—74 fires at $q=2$ (rate 0.0037), 115 at $q=5$ (rate 0.0057)—and the information is resolvable at $q=5$ ($I_D = 3.46 \times 10^{-4}$ nats, permutation $p = 0.001$) and marginal at $q=2$ (1.28×10^{-4} nats, $p = 0.053$). Raw sign-pair information on the same data is recorded in the artifacts as a scale diagnostic only; by Corollary 1 it is never detector evidence, and we do not report it as such.

Table 5: Information and net-value gates for the frozen cascade trigger. Kelly conversion $10,000 \cdot I_D$ (nats→bps); cost convention $C_{\text{rt}} = 10$ bps round-trip at unit size.

q	Triggers (of 20,126)	I_D (nats) [95% boot.]	Perm. p	Gross bound (bps)	W_D per-epoch (bps)	W_D per-trigger (bps)
2	74	1.28×10^{-4} $[0.21, 4.46] \times 10^{-4}$	0.053	1.28	−8.72	+1.25
5	115	3.46×10^{-4} $[0.38, 10.3] \times 10^{-4}$	0.001	3.46	−6.54	+3.41

Per-epoch: every decision epoch bears the full round-trip cost, $W_D = 10,000I_D - C_{\text{rt}}$. Per-trigger: cost is borne only on trigger epochs, $W_D = 10,000I_D - C_{\text{rt}} \times (\text{trigger rate})$; equivalently, per trigger event the gross bound is 349 bps ($q=2$) and 606 bps ($q=5$) against a 10 bps round-trip. Conditional forward returns on trigger epochs are negative on average (-1.7×10^{-4} at $q=2$, positive fraction 0.38), so the resolvable information is not a directional long signal; the bound is direction-agnostic.

The net-value column is the methodological finding of this subsection: under per-epoch attribution the trigger’s information is dominated by the 10 bps convention ($W_D = -8.72$ and -6.54 bps), while under per-trigger attribution the bound is positive (+1.25 and +3.41 bps per epoch). The two conventions have distinct semantics, not merely distinct arithmetic: per-epoch attribution is the coherent reading for an always-on state detector, and for a sparse trigger that implies action on 0.4–0.6% of epochs it is a deliberately conservative upper-cost convention; per-trigger attribution matches the trigger’s action semantics. The frozen tuple declared per-epoch attribution ex ante, so the gate input under the declared claim is *cost-dominated*, with the per-trigger sensitivity disclosed—and the lesson the certificate encodes is that the convention must match the declared action semantics when the tuple is frozen, because the verdict does not survive the substitution. Propagating the bootstrap intervals through Equation (5) bounds what either verdict can claim: per-epoch W_D at $q=2$ is resolved negative ($[-9.79, -5.54]$ bps) but at $q=5$ straddles zero ($[-9.62, +0.33]$ bps), so even the frozen-convention cost domination at the primary horizon is a point-estimate statement, not a 95%-resolved one; per-trigger W_D is positive across both intervals ($[+0.17, +4.42]$ and $[+0.32, +10.28]$ bps). Proposition 3 caps what any positive bound could license: an idealized fair-odds growth ceiling, not an attainable strategy—and the trigger-conditional forward returns (negative mean on

trigger epochs, positive fraction 0.38 at $q=2$) show the resolvable information is not a simple directional long signal. For a state-type detector that is always on, the two conventions coincide (Section 5), so the convention field is load-bearing precisely for sparse-trigger instruments.

4.6. A temporally disjoint 2026 window

Data from 2026-01-01 to 2026-05-29 (214,355 bars; 1,786 non-overlapping $W=120$ windows) were excluded from all development runs by a loader-level exclusion flag recorded in every development artifact. On the primary cell the cascade did not fire: observed $|\text{VR}(5) - 1| = 0.1575$ (mean 0.1835), median standardized statistic -0.3328 , asymptotic two-sided $p = 0.739$. The observed departure again exceeds the frozen exclusion bound (0.10) while lying below the demonstrated response threshold and on the negative side of the sign gate—the same disposition pattern as in-sample. We label this window a *temporally disjoint evaluation block* rather than a prospective holdout: the preregistration lineage (v3.0 amendment [dd44e7a](#), 2026-06-05; executed 2026-06-23) postdates the end of the window, so the separation is enforced by the recorded loader flag and freeze discipline, not by a third-party timestamp that predates the data. The distinction is exactly the kind of provenance fact a certificate exists to carry.

4.7. The completed certificate

Table 6 exhibits the certificate for the declared claim. This table is the paper’s central deliverable: every field is a measured quantity with a frozen artifact behind it, and the disposition follows mechanically from the precedence rule.

Non-detection has become a diagnostic vector rather than a generic failure: the cascade is a conservative instrument whose reference needs recentering, whose response threshold sits above the declared amplitudes, whose labels transport at one horizon but not another, and whose trigger information is real but convention-dependently valued. Each statement is bounded, artifact-backed, and independently actionable—and the next subsection acts on the first of them.

4.8. The diagnosis is actionable: recentering and the first admissible certificate

A certificate that only ever refuses is of limited use; the test of the protocol is whether its diagnoses repair the instrument. The size gate localized the fault precisely: the decision statistic (the median window-level Z) is referred to $N(0, 1)$, but under the dependence-preserving null it is centered near -0.005 ($q=2$) and -0.106 ($q=5$) with standard deviation ≈ 0.01 . We therefore replace the asymptotic reference with an empirical-null reference—studentizing the statistic by the null center and spread estimated from the $\delta=0.0005$ calibration cell and firing one-sidedly on positive departures—and re-evaluate every gate. Because each injection draw’s per-horizon statistic is stored in the frozen artifacts, the recentered rule is applied by exact offline re-thresholding of the same 104×200 draws: no new simulation, no raw data, fully reproducible, and by construction not a re-fit to a new outcome.

The effect is large and the arithmetic is auditable (Table 7, Figure 4). At $q=2, W=120$ the certified minimum detectable effect falls from $\text{cMDE} = 0.15$ under the frozen rule to 0.02 under the recentered rule—the detector now recovers $\delta = 0.02$ departures with probability one where before it recovered nothing below 0.15—while out-of-sample size (calibrated on half the null draws, evaluated on the other half) is 0.02. At $q=5$ the cMDE falls from 0.30 to 0.10 with out-of-sample size 0.07. The gain is genuine rather than an artifact of low Monte Carlo noise: the injection draws’ statistic has standard deviation ≈ 0.01 , matching the full-sample bootstrap standard deviation of the observed statistic (0.0088 at $q=2$, 0.0070 at $q=5$), so a $\delta = 0.02$ injection that shifts the statistic by several of these units is an effect a single deployment could resolve.

Applied to the observed market, the repaired instrument does not manufacture a detection. Recentered on its own dependence-preserving null, the observed BTCUSDT statistic sits at -1.9 null standard deviations at $q=2$ and -1.8 at $q=5$ —on the *negative* side, consistent with weak mean-reversion, which the positive-autocorrelation gate correctly declines to report as positive dependence. The difference from the frozen certificate is decisive for interpretation: the withhold is no longer underpowered. With $\text{cMDE} = 0.02$ at $q=2$, the repaired detector *excludes* positive short-horizon serial dependence above $\delta = 0.02$ with demonstrated

Table 6: Completed admissibility certificate: Holm-corrected VR cascade, BTCUSDT 2021–2025.

Field	Certified content
Claim tuple	Frozen cascade (q -grid $\{2, 5, 15, 60\}$, Holm 0.05, positive median gate); target: positive short-horizon serial dependence, economically read at $q=5$, $W=120$; scheme: calendar 1-min bars; signal family: positive AR(1) injections; null: centered circular block ($L=120$, post-freeze implementation choice); $\epsilon_g = \text{MDE} = 0.019748$; cost: 10 bps round-trip, unit size, per-epoch attribution. Freeze 1dc5c82 (2026-06-11), seed 43.
Power	96/96 preregistered cells silent at $\delta \leq 0.10$ (0/200 each, 95% upper bound 0.0183). Exploratory brackets: $\delta_{90}(q=2) \in (0.10, 0.15]$, $\delta_{90}(q=5, W=120) \in (0.20, 0.30]$. <i>Underpowered</i> for the declared amplitude family ($\delta \leq 0.10$).
Size	$\hat{\alpha}_D^* = 0/100$ $[0, 0.036]$ at both audited cells; interval excludes nominal 0.05; asymptotic reference mis-centered (null medians $-0.165/-0.389$, spans < 0.03). <i>Size-distorted, conservative direction</i> ; $B = 100$ resolution limit and single-null-family scope declared.
Transport	Certified at $q=2$ for clock-volume and volume-intrinsic; clock-intrinsic not certified under the preregistered margin (Holm 0.0682, interval inside ϵ_g but TOST not met); not certified at $q=5$ (clock-anchored discrepancies exceed ϵ_g); $q \geq 15$ event-time rows precision-limited. <i>Transport-uncertified</i> at the primary horizon.
Information/value	$I_D = 3.46 \times 10^{-4}$ nats at $q=5$ (perm. $p = 0.001$; plug-in bias order 0.25 bps); $W_D = -6.54$ bps per-epoch (frozen convention; interval $[-9.62, +0.33]$ straddles zero) vs. $+3.41$ bps per-trigger; <i>cost-dominated</i> under the frozen convention at the point estimate, convention- and interval-sensitivity disclosed.
Run integrity	All artifacts pin freeze/code commits and seeds; 2026 data loader-excluded in development (flag recorded); 8 post-freeze cells carry the exploratory provenance flag and are reported as exploratory.
Failure profile	Generated by the Section 3 precedence rule: (3) size-distorted (conservative) $\rightarrow$ (4) transport-uncertified ($q=5$) $\rightarrow$ (5) underpowered (declared amplitudes $\delta \leq 0.10$) $\rightarrow$ (6) cost-dominated (frozen convention, point estimate). Gates (1)–(2) pass.
Disposition	Withheld: size-distorted (conservative) , with the full profile above. The same evidence supports two bounded instrument statements: $q=2$ labels are transport-certified between calendar and volume bars and between volume and event-time bars, though not between calendar and event-time bars, and the cascade <i>cannot detect</i> departures $\delta \leq 0.10$ under this design (an instrument-scope statement; it admits no market claim about the existence or absence of dependence).

Table 7: Reference repair: certified minimum detectable effect before and after recentering the decision statistic on the empirical null ($W=120$; exact offline re-thresholding of the frozen injection draws).

q	cMDE, frozen $N(0, 1)$ rule	cMDE, recentered rule	δ_{50} , recentered	OOS size
2	0.15	0.02	0.02	0.02
5	0.30	0.10	0.05	0.07

cMDE is δ_{90} on the executed grid. The recentered cMDE at $q=2$ is stable across $W \in \{60, 120, 240\}$ (all 0.02); at $q=5$ it is 0.10 across W . The recentering constants are estimated post-freeze, so these are exploratory-grade cMDEs (Section 4.2); a preregistered confirmatory rerun with the constants frozen is required for a deployment certificate. Repair artifact: `backtest_results/reference_repair/recentered_reference_repair_20260710.json`.

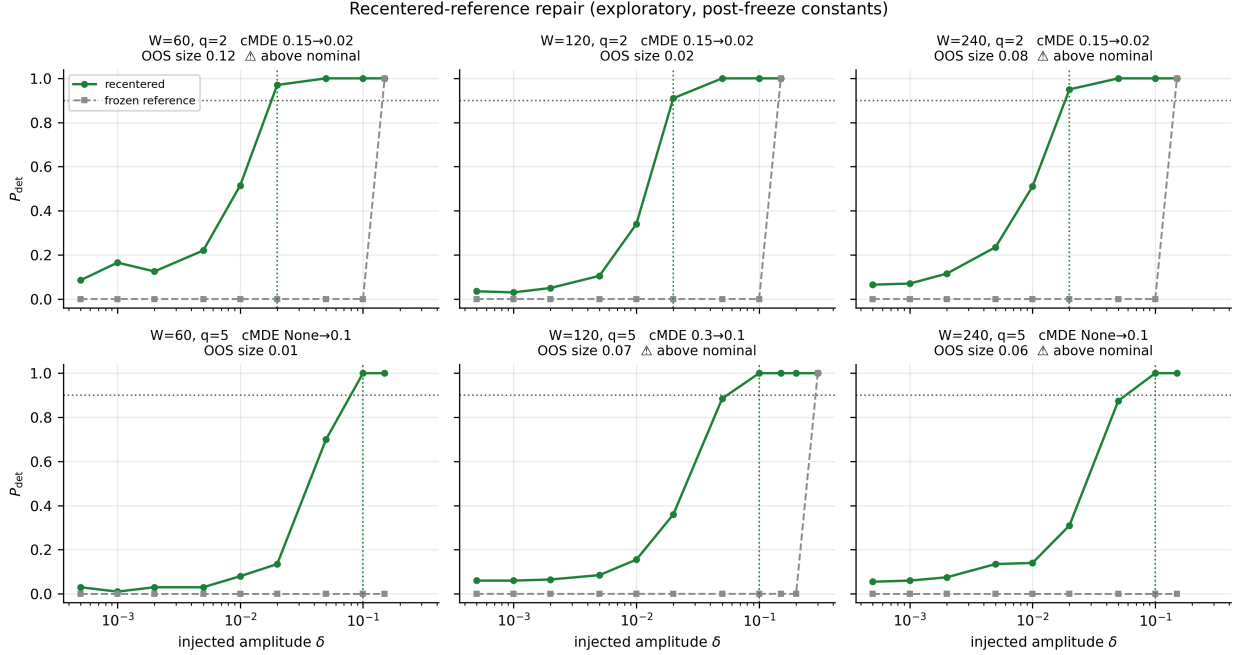


Figure 4: The frozen rule (red) recovers nothing below $\delta = 0.15$ ($q=2$) or 0.30 ($q=5$); recentering the decision statistic on the empirical null (blue) shifts the power curve left by roughly a decade, with out-of-sample size held near 5%. The certified minimum detectable effect falls into the preregistered amplitude grid, so the repaired instrument is adequately powered for the declared claim.

power, converting an uninterpretable non-result into a bounded, severe negative result in the sense of Mayo [8].

For the scoped scientific claim—positive short-horizon ($q=2$) serial dependence on calendar bars, without an economic target—the repaired instrument clears size (OOS 0.02) and power (cMDE = 0.02), and carries a transport gate that is met between calendar and volume bars but not between calendar and event-time bars, where the preregistered margin leaves the pair uncertified (Holm 0.0682, Table 4). Its disposition is therefore **admissible for the calendar-and-volume scheme scope**—the protocol’s first admission—with event-time transport explicitly outside the certified scope rather than quietly folded into it. This is a narrower admission than a fully transport-certified one would be, and the narrowing is recorded rather than absorbed: Table 8 exhibits the completed certificate. Two properties of this admission deserve emphasis. It was reached by repairing a diagnosed fault, not by relaxing a gate: the recentering is the standard empirical-null calibration the size gate itself prescribed. And admissible does not mean the market claim is affirmed—here the admitted certificate records a powered *exclusion*, which is precisely the kind of credible negative result that certification exists to license.

5. Three Detector Families Under One Protocol

Certification is detector-agnostic in a specific, testable sense: the same gates, run on different detector families, must produce dispositions for family-appropriate reasons rather than a single performance ranking. We demonstrate this on a harmonized three-family exhibit—rolling-quantile volatility detector, two-state Gaussian hidden-Markov filter [52], and the variance-ratio cascade—run on a native-frequency BTCUSDT block (2022-10-01 to 2022-12-31; 132,480 one-minute bars; q -units at native frequency; no downsampling), against a common two-state volatility-regime target with frozen validity gates. This exhibit is deliberately not a horse race: no agreement leaderboard is produced, and the protocol in fact *withholds* the pairwise agreement block because one family fails validity.

Table 8: First admissible certificate: recentered VR trigger, scoped claim (positive $q=2$ serial dependence, calendar bars, scientific target, BTCUSDT 2021–2025).

Field	Certified content
Claim tuple	Recentered VR trigger at $q=2$, $W=120$; target: positive short-horizon serial dependence (scientific, no economic target); calendar bars; positive AR(1) signal family; empirical-null reference from the $\delta=0.0005$ cell. Exploratory-grade (post-freeze recentering constants).
Size	OOS false-alarm 0.02 $[0, 0.05]$ at the declared window $W=120$; empirical-null reference replaces the mis-centered $N(0, 1)$. <i>Pass</i> at $W=120$. The recentering does not transport across the window grid: OOS size is 0.12 at $W=60$ and 0.08 at $W=240$, both above nominal. The $W=120$ scoping is therefore load-bearing, not incidental.
Power	cMDE = 0.02 (δ_{90}); $\delta_{50} = 0.02$; recovers the preregistered amplitude grid. <i>Pass</i> for $\delta \geq 0.02$.
Transport	Certified at $q=2$ for clock-volume and volume-intrinsic; clock-intrinsic not certified under the preregistered margin $\epsilon_g = 0.019748$ (Holm 0.0682; Table 4). <i>Pass within the calendar-and-volume scheme scope</i> ; event-time transport is outside the certified scope.
Disposition	Admissible for the scoped claim. Empirical content: positive $q=2$ serial dependence above $\delta = 0.02$ is <i>excluded</i> on BTCUSDT with power ≥ 0.9 (observed statistic -1.9 SD, negative side).
Scope note	The economic claim remains <i>cost-dominated</i> and the $q=5$ claim <i>transport-uncertified</i> (Table 6); admission is for the $q=2$ scientific claim only, at $W=120$, and across the calendar-and-volume scheme scope only. The recentering constants are post-freeze and exploratory-grade; a confirmatory rerun is required before this certificate could support deployment.

Table 9: Three families, one protocol (BTCUSDT 2022-Q4; two-state volatility-regime target; 89,222 overlapping-valid bars; cost convention 10 bps).

	Rolling quantile		Hidden Markov	VR cascade
Validity	Valid	(occupancy per-state non-overlap counts pass)	<i>Instrument-failed</i> : state 0 occupies 59 of 89,222 bars (0.07%); non-overlap count below frozen floor	Target gate: serial-dependence instrument, <i>target-mismatched</i> for a common volatility-state headline
Information (I_D , nats)	5.03×10^{-4}	$[3.47, 6.66] \times 10^{-4}$; perm. $p = 0.005$	Not interpreted (validity failed)	Not interpreted for this target
Net value	Gross 5.03 bps	-10 bps = -4.97 bps: <i>cost-dominated</i> . State label is always-on, so per-epoch and per-trigger conventions coincide	—	—
Missing gates	Power, size, transport not measured for this family	—	—	(Certified separately in Section 4)
Disposition	<i>Cost-dominated</i> (gate 6 fails on measured evidence, so the missing-estimand fallback does not apply; incomplete power/size/transport coverage recorded in the profile)	<i>Instrument-failed</i> (gate 1: read before any statistical gate)	<i>Target-mismatched</i> (gate 2) for this claim	

Pairwise agreement metrics (Cohen’s κ , adjusted Rand, NMI) were withheld by the protocol—the artifact’s pairwise block is empty—because a family member failed validity gates; run-level disposition: **instrument_failure**. The three-state variant of the task yields the same dispositions (rolling-quantile $I_D = 5.47 \times 10^{-4}$ nats, $p = 0.005$).

Three families, three dispositions, three different reasons—none of them a relative performance statement. The hidden-Markov failure is the precedence rule working as designed: a degenerate fit (one state absorbing 99.9% of bars) is an instrument failure to be reported *before* any information or agreement number is interpreted—and it is a repairable fit (constrained re-estimation or a longer window would likely populate both states), not a data point about hidden-Markov models in general. The rolling-quantile detector shows the intermediate state the rule produces when evidence is partial: its measured value gate fails ($W_D = -4.97$ bps), so the disposition is *cost-dominated* on measured evidence, while the unmeasured power, size, and transport gates are recorded as missing coverage rather than silently extrapolated. The cascade’s exclusion from a volatility-state headline is target eligibility, not failure. A leaderboard over these three streams would have converted an instrument failure, a target mismatch, and a cost-dominated value gate into a performance ranking; the protocol instead reports each for what it is.

6. What Certification Changes in Practice

Certification and the comparison toolkit answer different questions and compose in order. A reality-check, SPA test, or model confidence set [18, 20, 21] asks which candidate beats a benchmark, presupposing each label stream is a valid measurement of a common object; the certificate asks the prior question of whether that presupposition holds, and reports what no relative test does—mis-centered size, failed transport, convention-dependent value. Certification comes first (which outputs may enter), comparison second (among admitted outputs only).

For the empirical researcher. , the certificate converts negative results into bounded evidence with a stated power pedigree—the format that post-replication-crisis empirical finance explicitly calls for [10, 16]. A regime paper whose detector is certified underpowered below $\delta = 0.15$ cannot claim the market has no regimes; it can claim, defensibly, that departures above the threshold are excluded.

For the model validator. , the certificate is a model-risk artifact in the sense of Board of Governors of the Federal Reserve System and Office of the Comptroller of the Currency [3]: a pre-use validation with explicit expiry and recalibration triggers, extending VaR-style backtesting [1, 2] to label-producing models that current backtesting practice does not cover.

For the trading researcher. , the value gate separates three routinely conflated objects: information in the market path (a ceiling, by Corollary 1), information in the detector output (the estimand), and value net of a declared cost convention (the gate). The demonstrated convention-dependence is a caution against both optimistic and pessimistic shortcuts: charging every epoch a full round-trip buried a positive per-trigger bound; per-trigger accounting, taken alone, would overstate what an upper bound licenses. The certificate’s answer is not a number but a frozen convention plus a disclosed sensitivity.

7. Domain of Validity, Limitations, and Sequential Extension

Every empirical statement in this paper is bounded by Table 10, and we make the boundaries explicit rather than relying on generalization by analogy. Three limitations deserve emphasis. First, the size diagnostic runs at $B = 100$: it demonstrates the gate and bounds the false-alarm rate below 3.6%, but a deployment certificate requires a pre-declared budget sized to its resolution target, across multiple null families. Second, power, size, and transport were measured for one detector family; the three-family exhibit shows the protocol’s refusal states on other families, not their full certificates—completing a non-VR certificate is the natural next study, and the *incomplete* disposition is the protocol saying so itself. Third, the injection design perturbs the empirical path with a single signal family (positive AR(1) calibrated to $|\text{VR}(q) - 1|$); a claim about a different signal shape requires refreezing s . A specification-curve presentation of the certificate across reasonable tuple variations [53]—and multi-asset extensions run as registered reports—are the two designed-in growth paths for the evidence base.

Table 10: Domain of validity of the empirical claims.

Dimension	Certified scope
Detector	Frozen Holm-corrected LM variance-ratio cascade with positive median gate (Section 4); rolling-quantile and HMM families touched only by the gates shown in Table 9.
Asset/venue	BTCUSDT Binance USD-M perpetual futures; ETHUSDT same-venue replication in Appendix B; EUR/USD used for cost-term instantiation only (Appendix C).
Period	Development 2021–2025 (2021-05-29 to 2025-12-31); disjoint 2026-01-01 to 2026-05-29 evaluation block.
Frequency/schemes	One-minute base bars; calendar, volume, event-time constructions per the frozen calibration table.
Horizons	$q \in \{2, 5, 15, 60\}$ bars; windows $W \in \{60, 120, 240\}$.
Signal family	Positive AR(1) injections targeting $ \text{VR}(q) - 1 \in [0.0005, 0.30]$ as executed.
Nulls	Centered circular block bootstrap, block 120, $B = 100$ (resolution-limited as stated).
Costs	10 bps all-in round-trip at unit size; per-epoch attribution frozen, per-trigger sensitivity disclosed; EUR/USD spread evidence for cost-term instantiation.
Claims excluded	Any profitability claim; any market-efficiency verdict; any cross-asset or cross-family generalization beyond the rows above.

Sequential extension. Production use needs a monitoring form of the protocol: admit a detector when rolling evidence first clears the gates, revoke when it degrades. A window-by-window p-value monitor is the wrong instrument—continuous looking with optional stopping invalidates its nominal guarantees, a problem known since sequential monitoring of structural change [54]. Proposition 5 supplies the correct instrument: monitoring each gate by an e-process makes admission anytime-valid, so the false-admission probability is bounded over an unbounded horizon under optional stopping, with no error budget split across gates or looks. This is a stronger and simpler guarantee than the fixed max-statistic budget of the earlier prototype. The remaining step is empirical: the repository’s prototype state machine replays a *scripted* schedule rather than calibrated e-processes, which under this paper’s own rules is not admissible evidence, so we report the guarantee and its construction but withhold a sequential empirical demonstration until the monitor is rebuilt on the betting e-process of Section 3.4 and certified against replayed history.

8. Conclusion

Regime detectors are measurement instruments, and this paper’s proposal is that they be treated with an instrument’s discipline: no output enters inference or allocation until its power, false-alarm behavior, transportability, and net information value have been measured for one frozen claim. The deliverable is the certificate, and a withheld admission carries as much usable information as a granted one, because it names the failed gate.

The demonstration certifies a variance-ratio cascade on five years of one-minute Bitcoin futures and withholds admission, for measured reasons: the rule’s empirical false-alarm rate sits significantly below its nominal level, with a mis-centered asymptotic reference—a size distortion in the conservative direction; a preregistered 96-cell grid shows the cascade cannot detect departures at any amplitude up to 0.10, with the response threshold bracketed at $(0.10, 0.15]$ at $q=2$ and $(0.20, 0.30]$ at $q=5$; labels transport across calendar, volume, and event-time bars at $q=2$ but not at the primary horizon; and trigger information, resolvable at $q=5$, has a net value whose sign depends on a cost-attribution convention that the certificate therefore freezes and discloses. A three-family exhibit shows the same protocol returning instrument-failed, target-mismatched, and cost-dominated dispositions rather than a leaderboard.

None of this is a verdict on Bitcoin’s efficiency, on variance-ratio tests, or on any trading strategy. It is a worked template for a validation step that regime-detection practice currently skips: bounded claims, measured power, disclosed provenance, and refusal as a reportable outcome.

Appendix A. Derivations

Proposition 1.. Conditional on the empirical path X , the injection design, and the frozen rule, each Monte Carlo draw applies independent injection randomness to the same measurable event $\{D_\theta(T_g(J_{\delta,s}(X))) \text{ fires}\}$; draw outcomes are i.i.d. Bernoulli(P_{det}) by construction of the simulator, so the sample fire rate is the maximum-likelihood estimator of P_{det} and Clopper–Pearson intervals are exact at any N_{mc} . The conditionality is irreducible: nothing licenses extrapolation beyond the declared path, family, and grid.

Proposition 2.. Immediate from the definition of certification: a transported claim requires certified equivalence of the declared statistics; absent certification, the transported claim has no measured basis, which is the withholding rule. The content is operational, not probabilistic.

Proposition 3 and Corollary 1.. The fair-odds log-optimal bound gives expected log-growth advantage at most $I(D; Y)$ nats per decision [32, 33, 35]; converting to basis points and subtracting the declared cost under the declared attribution convention yields W_D ; non-positive W_D excludes detector-only use under that convention, and positivity remains an upper bound because the Kelly bettor is idealized (perfect odds, no constraint, no estimation error). The corollary is the data-processing inequality applied to the causal map $X_{\leq t}^g \mapsto D_t^g$.

Proposition 4.. Under $H_0 = \bigcup_k H_0^k$ there is a gate k^* with $H_0^{k^*}$ true. Admission requires $E_{k^*} \geq 1/\alpha$, so by Markov’s inequality and the e-value property, $\Pr_{H_0}(\text{admit}) \leq \Pr_{H_0^{k^*}}(E_{k^*} \geq 1/\alpha) \leq \alpha \mathbb{E}_{H_0^{k^*}}[E_{k^*}] \leq \alpha$. No union over gates appears because admission is a conjunction; the bound is controlled by the single true null. That the product of fold-wise e-values and the average of arbitrary e-values are themselves e-values are, respectively, the tower property and linearity of expectation [45].

Proposition 5.. Under H_0 some gate k^* fails at all t , so $(E_t^{k^*})$ is a nonnegative supermartingale with $\mathbb{E}[E_0^{k^*}] \leq 1$. Ville’s inequality gives $\Pr(\exists t : E_t^{k^*} \geq 1/\alpha) \leq \alpha \mathbb{E}[E_0^{k^*}] \leq \alpha$. Admission at any t requires $E_t^{k^*} \geq 1/\alpha$, hence $\Pr(\exists t : \text{admit}) \leq \alpha$; the bound is uniform over t and so survives any data-dependent stopping rule [48].

Appendix B. ETHUSDT External Replication

The full workflow was rerun on ETHUSDT Binance USD-M perpetual futures at one-minute frequency, same effective window as BTCUSDT (2021-05-29 to 2025-12-31, $n = 2,415,148$ bars) and same disjoint 2026 block (through 2026-05-29, $n = 214,355$ bars). The run changes the asset symbol only; BTC injection artifacts are not reused as ETH calibration data, and no ETH-specific injection grid or empirical-size run exists (`eth_specific_injection_grid: false`), so the ETH certificate is *incomplete* by its own provenance—exactly the disposition the protocol requires rather than a silent extrapolation.

Measured gates replicate the qualitative BTC pattern with instructive differences. Primary cell: median $|\text{VR}(5) - 1| = 0.1609$ (mean 0.1832), median standardized statistic -0.3981 , asymptotic $p = 0.691$, cascade silent—large departure, wrong sign for the positive gate, below the demonstrated response threshold. Transport (same margin, same 12-row family): certified at $q=2$ for clock-volume ($\hat{\Delta} = 0.0035$, Holm 6.2×10^{-42}) and volume-intrinsic (0.0094, Holm 0.014) but *not* clock-intrinsic (0.0129, Holm 0.154); certified at $q=5$ for clock-volume (0.0110, Holm 0.0014) and at $q=60$ for clock-volume (0.0050, Holm 0.0043); all event-time pairs at $q \geq 5$ fail, with the $q \geq 15$ rows precision-limited as on BTC. The certified set differs from BTCUSDT’s—transport certification is an asset-specific measurement, not a detector constant, which is the point of gating it. Persistence diagnostics reject the dependence-preserving null (KS = 0.261, bootstrap $p < 0.001$), and the disjoint 2026 block repeats the primary-cell disposition ($|\text{VR}(5) - 1| = 0.1568$, median $Z = -0.3126$, $p = 0.755$, 1,786 windows). Raw sign-pair information bounds (9.2 bps gross at $q=5$) are recorded in the artifact as scale diagnostics only and are not detector evidence under Corollary 1.

Appendix C. EUR/USD Cost-Term Instantiation

Daily reference FX rates contain no executable quotes, so using them for the cost gate would force a frictionless exception to Equation (5). The repository instead instantiates the cost term from free HistData Generic ASCII tick quotes with bid, ask, and volume: 56 monthly files, 113,650,998 EURUSD ticks, aggregated to 1,659,616 one-minute last-quote bid/ask bars over a BTC/ETH-matched window (2021-05-29 19:32 UTC to 2025-12-31 23:59 UTC; first available quote 2021-05-30 17:00, last 2025-12-31 16:58, reflecting FX session hours). Median observed spread: 3.0×10^{-5} (0.284 bps of mid); mean 0.417 bps; matched-span maximum 30.95 bps. This appendix is deliberately narrow: it shows the certificate’s cost term can be instantiated on a traditional asset without paid institutional data, and it is not a second-asset detector certificate (power, size, transport, and information gates were not run under an FX claim tuple).

Appendix D. Reproducibility and Legacy Naming

The replication package separates method, claim boundaries, code, tests, and frozen evidence. Three reproduction levels are supported: (i) rebuild the manuscript and check every reported value against the frozen artifacts via the claim-traceability ledger; (ii) run the focused verification tests against the included scripts and artifacts; (iii) regenerate comparable artifacts from independently obtained raw data using the documented commands. Raw venue and vendor data are not redistributed.

The package separates the manuscript and figures (`paper/`), the claim boundary (`CLAIMS.md`, `CLAIM_TRACEABILITY.md`, `ARTIFACT_LEDGER.md`), the method implementation (`scripts/wp1/`, `src/strategies/vol_regime_switch/`), the frozen evidence (`data/injection_runs/`, 104 power cells; `backtest_results/` for the size, transport, information, disjoint-window, replication, benchmark, reference-repair, and cost artifacts), and the verification tests (`tests/wp1/`); raw Binance USD-M and HistData inputs are identified in the runbook but not redistributed.

Integrity conventions: the detector and claim tuple are frozen before interpretation (preregistration commits recorded in every artifact); the 2026 block is excluded from development by a recorded loader flag; seeds and bootstrap settings are stored in scripts or artifacts; derived artifacts are immutable evidence anchors; post-freeze evidence carries an explicit exploratory flag that the manuscript must disclose wherever such evidence is displayed; and regeneration writes new output paths unless manuscript and ledger are updated together.

Legacy naming. The protocol was developed in the repository under the working name GCDE (“gauge-calibrated detector exclusion”), and frozen artifact paths and keys retain that vocabulary (`gauge_invariance/`, `gauge_scope`, `thermodynamic_bound/`). The correspondence is: gauge $\rightarrow$ sampling scheme / market clock; gauge defect $\rightarrow$ transport discrepancy Δ_D ; available decision work $\rightarrow$ net information value W_D ; exclusion $\rightarrow$ withheld admission; failure fingerprint $\rightarrow$ failure profile. Artifact keys are quoted verbatim where they appear so that ledger lookups remain exact.

Data and Code Availability

The demonstration uses locally archived Binance USD-M BTCUSDT and ETHUSDT one-minute perpetual futures data, a free matched-span HistData EURUSD bid/ask sample, and frozen derived artifacts in the project repository. Principal scripts: `scripts/wp1/`; verification tests: `tests/wp1/`; frozen artifacts: `data/injection_runs/` and `backtest_results/`; evidence map: the artifact ledger and claim traceability files. The public replication repository mirrors the paper package structure. Raw market-data redistribution is subject to the original venue and vendor terms.

CRedit Authorship Contribution Statement

Diego Urdaneta Achong: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Writing — original draft, Visualization, Project administration. **Claudio Martel Flores:** Software, Validation, Formal analysis, Writing — review & editing. **Melis Boztosun:** Methodology,

Software, Validation, Writing — review & editing. **Everett Zhang:** Investigation, Resources, Writing — review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of Generative AI and AI-Assisted Technologies

During the preparation of this work the authors used AI-assisted coding and planning tools (Cursor and Claude) for code debugging, implementation support, and drafting of internal planning and review documentation retained in the project repository. All research design decisions, pre-registered specifications, threshold choices, and disposition verdicts were made and ratified by the authors; no analytical result, statistic, or verdict reported in this paper was generated or adjudicated by an AI system. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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